

Hiding in Plain Sight: Bias, Reform, and Notch Upgrading

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Abstract

This paper studies how scale architecture shapes evaluator bias. In international figure skating, the same nine-judge panel scores each performance simultaneously on a rule-coded objective channel and a subjective channel. Across 1,012,339 item-level scores over ten seasons, I find that compatriot bias is nearly three times larger on the objective channel, reversing the canonical pattern that bias concentrates on subjective scoring. A within-performance triple-difference around two ISU reforms identifies the mechanism. The 2018/19 reform narrowed the rule-defensibility region around each integer boundary and attenuated bias by 42%. The 2022/23 reform consolidated subjective components without changing boundary geometry; average bias was statistically unchanged, but doubled at the perfect-10 ceiling. Reforms that change boundary architecture attenuate bias; reforms that leave it intact relocate it.

JEL Codes: J71, L83, D91.

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1 Introduction

For decades, organizations have responded to evaluator bias by reforming the rules: more transparent procedures, more structured scoring scales, more rule-coded categories (Goldin and Rouse, 2000). The reform paradigm rests on a shared assumption: bias finds room mainly in subjective judgment, where evaluators have the most discretion (Prendergast and Topel, 1996; Zitzewitz, 2006; Garicano et al., 2005; Hoffman et al., 2018). Audits have followed this logic, monitoring the subjective margins where bias is theorized to concentrate. In labor markets, resume audits document large callback gaps for identical applications differing only in name-signaled race (Bertrand and Mullainathan, 2004). In NBA refereeing, an early audit documented racial-match-driven foul calls (Price and Wolfers, 2010); the bias on that audited margin attenuated after the study’s publicity (Pope et al., 2018). But a single-channel audit cannot test whether bias was eliminated or relocated to unaudited margins. Yet the assumption has rarely been tested in evaluation systems where structured “objective” scoring coexists with subjective rating in the same evaluation moment. This paper asks whether the bias that audits no longer find on subjective channels has truly been eliminated, or whether it has simply been hiding on the “objective” channel that conventional wisdom treats as transparent and beyond suspicion.

Answering this question has been hard because most evaluation systems offer only a single rating channel for audit to observe. A student receives one numerical score; a credit applicant receives one risk grade; a job candidate receives one ranking. When audits detect bias on this single channel and reform later attenuates it, the natural conclusion is that the reform reduced bias. The conclusion only follows, however, if the audit has examined every channel where bias could plausibly operate, which most systems do not allow; even where they do, we cannot directly observe the evaluator’s decision process at the moment of rating. An ideal test of this question would split evaluation into two structurally distinct rating channels, one subjective and one objective (Baker et al., 1994), that the same evaluator scores simultaneously for the same case. Comparing the two scores would tell us where bias actually comes from.

International figure skating provides field data for this test. Every performance is scored on two

structurally distinct rating channels by the same nine-judge panel. The Grade of Execution (GOE) records each technical element on a coarse integer scale of ± 5 . The Program Component Score (PCS) records holistic skating qualities on a fine 0.25-point scale spanning forty levels ([International Skating Union, 2018](#)). The architecture is well suited to the test on four counts. The same panel scores both channels for the same skater in the same competition, so skater quality, panel composition, and event conditions are differenced out within each performance. The channels differ in objectivity: GOE is scored against explicit technical rules, while PCS captures holistic artistic judgment. Judge and skater nationalities have been publicly posted since 2016, supplying the in-group signal we use to detect bias: compatriot favoritism. The judges are ISU-licensed professionals operating across a tiered competition circuit, where scores affect athlete rankings, prize money, and qualification through a hierarchy that culminates in World Championships and the Olympics.

For identification, I exploit a 2018/19 ISU reform that changed the scale architecture of one channel and left the other unchanged. The reform widened the GOE range from ± 3 to ± 5 , expanding seven integer levels to eleven and narrowing the gap between adjacent values; the 0.25-point PCS scale was left untouched ([International Skating Union, 2018](#)). The GOE expansion was framed as a technical adjustment to permit finer execution differentiation, not as a bias-reduction reform. In conventional reform-paradigm terms, this should make the objective channel more objective and narrow the space where bias could operate. Within each performance, the same compatriot judge scores both the reformed GOE and the unchanged PCS, so the untouched PCS serves as a within-performance counterfactual for the reformed GOE. The full sample covers over a million individual scoring decisions: 543 judges from 55 nations and 1,027 skaters from 64 nations across the universe of senior international ISU competitions over ten seasons (2016/17–2025/26).

Three findings emerge. The canonical pattern that bias concentrates on subjective scoring ([Zitzewitz, 2006](#)) no longer holds: compatriot bias is nearly three times larger on the more objective GOE channel than on the more subjective PCS channel. The 2018/19 architectural reform cut the GOE compatriot premium by 42%, operating through what I call *notch upgrading*¹: at integer-boundary

¹I use *notch* and *boundary* interchangeably to refer to the zone between two adjacent discrete score values, drawing on the public-finance literature on discontinuous payoff structures ([Kleven and Waseem, 2013](#)). [Duggan and Levitt](#)

midpoints, same-nationality judges round up 20.6 percentage points more often than panel peers pre-reform, falling to 14.7 points after the integer gaps narrow.

The 2022/23 PCS component reduction, which consolidated five components into three but left the 0.25-point boundary grid intact, produced no comparable attenuation. The PCS same-nationality premium was unchanged at the average. On the PCS scale, the same-nationality upgrade premium was 45.5 percentage points across the three top-end midpoints pre-reform and roughly doubled at the perfect-10 boundary (9.75/10.00) after the reform, the only architectural boundary on a fine-grid channel. Bias migrated to the maximally rule-defensible boundary that the reform left standing rather than dissipating. The two reforms together expose a common mechanism: bias hides in plain sight on the channel the audit logic certifies as objective, and reforms that leave boundary geometry intact relocate bias rather than remove it.

The architectural mechanism is this. At a boundary between two adjacent rule-coded scores, both scores are defensible: a judge who picks the higher cannot be cited for a specific rule violation. The zone where this dual defensibility holds is the *rule-defensibility region* around the boundary, and bias accordingly settles there: the cost of defending a biased choice is low. Bias also depends on the architectural payoff, largest on elements where the upgrade is worth more in raw points. The 2018/19 reform reshapes the GOE architecture by inserting four new integer levels; the boundary ambiguity shrinks and the bias attenuates. The same mechanism produces a 45.5-point notch-upgrading premium across PCS's top-end midpoints, the region surrounding its one architectural boundary at the perfect-10 ceiling. This within-scale boundary mechanism differs from scale-design interventions that operate by removing extreme values that activate stereotypes ([Rivera and Tilcsik, 2019](#)) or by dichotomizing the rating to eliminate the intermediate gradations through which subtle bias operates ([Botelho et al., 2025](#)). The mechanism is portable: any structured rating where adjacent levels can be defended under the same rules should host it. This bias does not require subjectivity in the channel, and need not be conscious: where the rule book certifies two adjacent scores as defensible, the architecture lets motivated reasoning ([Kunda, 1990](#); [Dana et al.,](#)

(2002) document the analogous agent-side response in sport: sumo wrestlers near the rank-promotion threshold win at sharply elevated rates.

2007) prefer one without violating any rule and without recognizing the preference as bias (Norton et al., 2004).

Relation to literature. First, the literature on evaluator bias in elite professional evaluation has historically located bias on subjective margins where evaluators have the most discretion: in resume callbacks (Bertrand and Mullainathan, 2004), NBA refereeing (Price and Wolfers, 2010; Pope et al., 2018), baseball strike-zone calls amplified at the edge-zone where rules grant umpires discretion (Parsons et al., 2011), soccer injury-time decisions (Garicano et al., 2005), and winter-sport judging (Krumer et al., 2022), most canonically figure skating's pre-2003 6.0 system, where Zitzewitz (2006) documents nationalistic bias concentrated on the artistic-impression channel. Zitzewitz (2014) examines the post-2003 anonymization reform and finds modest, statistically insignificant changes in compatriot bias, with bias appearing larger on the technical-merit channel under the new code-of-points than under 6.0, an aside that pre-figures the migration this paper documents. Three contributions distinguish this paper. The first traces a twenty-year migration of compatriot bias. Under the rule-coded system that replaced the 6.0 in 2003-04, I document that compatriot bias is nearly three times larger on the more objective rule-coded channel than on the subjective one. Reform aimed at the subjective margin has not eliminated bias; it has migrated to the rule-coded channel that audits certify as objective. The second is a mechanism refinement: where Parsons et al. (2011) and Garicano et al. (2005) locate bias amplification in regions where rules grant evaluators discretion, I show that bias also amplifies at architectural boundaries where the rules certify two adjacent values as jointly defensible. Defensibility is a sharper mechanism than discretion, operating inside rule-coded channels that discretion alone does not enter. Where these prior papers identify bias at a single architectural point, I document the architectural geography across a full scoring system. The third is a methodological diagnosis: single-channel audits cannot distinguish bias eliminated from bias relocated, and post-reform single-channel measurements (Pope et al., 2018; Zitzewitz, 2014) can equally reflect bias migration to a rule-defensible margin that audits do not reach. The within-performance dual-channel design here resolves this distinction.

Second, the literature on the architecture of evaluation studies how rating-system design shapes bias. Each documented bias-attenuating reform operates on a different ingredient of the bias process: removing demographic signals (Goldin and Rouse, 2000), removing scale extremes that activate stereotypes (Rivera and Tilcsik, 2019), removing intermediate gradations through dichotomization (Botelho et al., 2025), shifting from separate to joint evaluation (Bohnet et al., 2016), and replacing human evaluators with algorithms (Kleinberg et al., 2018). Theoretical foundations for combining subjective and objective measures are developed in Baker et al. (1994); Prendergast and Topel (1996); Hoffman et al. (2018), and rankings reactivity in public measurement is studied by Espeland and Sauder (2007). I add a different ingredient: the rule-defensibility region around each architectural boundary in a discrete rating scale. The 2018/19 GOE expansion from seven to eleven integer levels narrows this region around each boundary and attenuates compatriot bias by 42%. The 2022/23 PCS reduction leaves the boundary geometry intact, and bias relocates to the perfect-10 ceiling rather than dissipating. The relationship between scale design and bias is therefore mechanism-contingent: appropriate redesign depends on diagnosing which ingredient is active. The mechanism connects to the public-finance literature on threshold behavior, which documents agent-side bunching at thresholds with payoff consequences (Kleven and Waseem, 2013; Mayo, 2025; Pope and Simonsohn, 2011; Duggan and Levitt, 2002); my notch-upgrading test transfers that pull to the evaluator. It is distinct from existing evaluator-side notch literature on test-score manipulation (Diamond and Persson, 2016; Dee et al., 2019) in motivation: in-group preference at architectural boundaries rather than altruistic boost at consequential pass/fail thresholds.

Third, the literature on motivated reasoning, plausible deniability, and the sources of discrimination supplies the psychological substrate for the architectural mechanism. Motivated reasoning theory establishes that biased preferences require constructed justifications to be acted upon (Kunda, 1990), with moral wiggle room (Dana et al., 2007) and justification-suppression dynamics (Crandall and Eshleman, 2003; Norton et al., 2004) supplying the individual-level mechanisms. Pikulina and Ferreira (2026) formalizes *subtle discrimination* as biased acts that cannot be objectively ascertained as discriminatory, with plausible deniability as the defining feature. Field evidence shows

that meritocratic framing licenses bias rather than reducing it (Castilla and Benard, 2010), and debiasing information updates beliefs but leaves taste-based discrimination intact (Montoya et al., 2026); both find that interventions miss when the locus of bias is wrong. The framework of Bohren et al. (2025a) decomposes discrimination into three sources: taste-based animus, accurate statistical inference, and inaccurate beliefs about group quality. I make three contributions to this literature. The first is an institutional analog: rule architectures themselves can certify two adjacent scores as defensible, supplying the institutional form of moral wiggle room. This operates without requiring the channel to be subjective and without requiring conscious intent on the part of the evaluator. The second is upstream of the source decomposition: the architectural mechanism is consistent with each of the three sources of discrimination, since architectural cover supplies what any form of motivated choice requires. The third extends “wrong-locus” findings from organizational culture (Castilla and Benard, 2010) and intervention design (Montoya et al., 2026) to evaluation architecture: reforms that target subjective channels miss bias that has migrated to rule-defensible regions on objective channels.

2 Institutional Setting

International figure skating offers a laboratory-like environment for studying how evaluation architecture shapes bias: two scoring domains with different boundary granularity coexist within each performance, scored simultaneously by the same nine-judge panel, with publicly posted judge nationalities since 2016. Nationalistic bias in winter-sports judging is documented in figure skating (Zitzewitz, 2006, 2014) and ski jumping (Krumer et al., 2022), but the architectural mechanism through which it operates has not been identified.

Since the 2004/05 season, the International Skating Union (ISU) has evaluated competitions using the ISU Judging System (IJS), which replaced the older 6.0 ordinal system after criticism of opaque judging and of vote trading (Appendix A provides additional institutional detail; both appendices are available online). A panel of nine judges evaluates each performance along two

dimensions: the Grade of Execution (GOE) and the Program Component Score (PCS).

Grade of Execution. GOE measures the technical quality of individual elements such as jumps, spins, and step sequences. Each element carries a base value set by the ISU technical committee, and each judge adjusts this value by assigning an integer grade that reflects execution quality. ISU Communication 2168 governs GOE marking through bullet-point criteria: a +1 grade requires one positive feature (e.g., “good height,” “good take-off and landing,” “effortless throughout”), a +2 requires two, and so on, with three of five mandatory bullets required for +4 or +5 ([International Skating Union, 2018](#)). Before the 2018/19 season, the GOE scale ranged from −3 to +3 (seven integer levels). Starting in 2018/19, the ISU expanded the scale to −5 to +5 (eleven levels) to permit finer differentiation of execution quality.

Program Component Score. PCS captures broader artistic and skating qualities. Judges rate each component on a scale from .25 to 10.00 in 0.25-point increments, yielding 40 possible values. Before the 2022/23 season, five components were scored; the ISU then consolidated these into three (Skating Skills, Composition, and Presentation) ([International Skating Union, 2022](#)). The PCS scale was unaffected by the 2018/19 GOE expansion.

Aggregation. For each scored item, the ISU drops the highest and lowest of the nine judges’ marks and averages the remaining seven. The skater’s segment score equals the sum of these trimmed means across all items. Final standings are determined by the combined segment scores of the short program and free skate.

Panel assignment. Judges are assigned to competition panels by draw. At ISU Championships and the Olympic Winter Games, a pool of 13 judges is drawn for each discipline, and 9 are selected by lot for each segment. No ISU member country may have more than one judge on a panel. This quasi-random assignment mitigates concerns about systematic selection of judges into panels with compatriot skaters.

Judge anonymity. Since the 2016/17 season, the ISU has published individual judges’ marks alongside judge names on each competition’s protocol sheet. [Zitzewitz \(2014\)](#) studies the earlier anonymized period. Throughout the sample period (2016/17–2025/26), judge identities are fully

public.

Two features of this setting are central to the empirical strategy. First, the two scoring domains differ in boundary structure. GOE uses a coarse integer scale with wide gaps between adjacent values; PCS uses a fine quasi-continuous scale with narrow 0.25-point increments. This contrast permits a within-performance, within-panel comparison of bias across domains that differ in joint-level exploitability.

Second, the 2018/19 GOE expansion supplies an exogenous change to the boundary structure of one domain but not the other. The expansion increased the number of integer levels from seven to eleven, narrowing the distance between adjacent values. If bias operates through strategic upgrading at score boundaries, a finer scale should reduce the return to notch upgrading and attenuate bias. PCS, which was unaffected by the expansion, serves as a within-performance control.

A third feature offers a supplementary test. The 2022/23 PCS reduction consolidated five components into three but left the 0.25-point scoring scale unchanged. Because the reduction altered the number of scored items without changing boundary structure, the theory predicts no attenuation of PCS bias. Section 5.2 tests both reforms in a single saturated specification.

3 Data

3.1 Sample

I collect individual judge-level scores from official ISU protocol sheets for all major figure skating competitions over 10 seasons (2016/17 through 2025/26). The data cover the Olympic Winter Games, World Championships, Grand Prix Final, Grand Prix series, European Championships, and Four Continents Championships. For each performance, the ISU publicly reports each judge's score on each scored item, along with judge and skater nationalities. I compile the dataset by parsing the PDF protocol sheets published on the ISU results website (<https://www.isu.org>).

The sample restricts to ISU-sanctioned senior international competitions where panel assignment follows the draw procedure described in Section 2. I exclude Challenger Series events (dif-

ferent officiating regime), the Junior Grand Prix circuit (different element values and judge pool), national championships (federation-specific rules), exhibition galas (unscored), team events, and observations where either nationality is unknown. The senior international circuit keeps institutional rules comparable across competitions and preserves the draw-based panel assignment central to identification.

The unit of observation is a judge-item pair within a skating segment. Table 1 reports summary statistics. The final sample contains 1,012,339 item-level observations (693,885 GOE and 318,454 PCS), scored by 543 judges across 44 competitions and 7,372 athlete-segments. Same-nationality pairs make up 7.8% of the sample.

Pre-registration. This analysis was not pre-registered. The interaction between scale granularity and compatriot favoritism became the object of study during initial exploratory work on the 2016/17–2017/18 judging panels, and the 2018/19 GOE expansion was identified as the relevant source of variation during that exploration. The notch-upgrading mechanism, the triple-difference specification, and the scale-invariant upgrade indicator were specified *ex ante* of viewing the post-expansion seasons, but the analytical apparatus was not lodged publicly in advance. Two analyses are post-hoc and flagged accordingly. The PCS top-end midpoint restriction {9.375, 9.625, 9.875} in Appendix B.5 was specified after observing the GOE notch-upgrading pattern, and the single-boundary 9.875 estimate was added in response to the redistribution finding. The mid-scale placebo (Table B5) was developed to test these analyses against the alternative that the effect is a 10.00-ceiling artifact rather than an architectural-boundary phenomenon. The paper reports its full set of pre-planned tests together with the post-hoc analyses and the robustness sweep developed in response to external review.

3.2 Measurement

For each judge-item pair, I compute the normalized score deviation: the focal judge’s score minus the leave-one-out panel mean (the average of the other eight judges), divided by the item-level

standard deviation computed separately in the pre- and post-2018 eras (the era-specific normalization defined in Section 4.2). The leave-one-out construction ensures that the deviation numerator is mechanically independent of the focal judge’s same-nationality status.

The key independent variable equals one when the judge and skater share the same nationality. Panel B of Table 1 previews the pattern. In GOE, same-nationality judges score .052 standard deviations above the panel mean; different-nationality judges score .004 below it, yielding a raw gap of .056. In PCS, the gap is .022. The raw same-nationality gap is more than twice as large on the more objective of the two scoring domains, a pattern that reverses the canonical claim that bias concentrates on subjective scoring (Zitzewitz, 2006) and motivates the analysis that follows.

TABLE 1
SUMMARY STATISTICS

<i>Panel A: Sample overview</i>		
Total item-level observations	1,012,339	
Unique judges	543	
Unique competitions	44	
Unique athlete-segments	7,372	
Seasons	10	
Same-nationality share (%)	7.8	
<i>Panel B: Score characteristics by domain</i>		
	GOE	PCS
Item-level observations	693,885	318,454
Mean score	.95	7.47
SD of score	2.10	1.16
Mean normalized deviation	.127	.032
Same-nationality share (%)	7.9	7.8
Mean normalized deviation (same nationality)	.052	.020
Mean normalized deviation (diff. nationality)	-.004	-.002
<i>Panel C: Institutional design</i>		
	GOE	PCS
Score type	Integer	.25 increments
Pre-2018 scale	±3 (7 levels)	0–10 (41 levels)
Post-2018 scale	±5 (11 levels)	0–10 (41 levels)
Reform	2018/19 expansion	2022/23 reduction

Note.— Unit of observation: judge × scoring item within an athlete-segment. GOE (Grade of Execution) evaluates technical elements on a discrete integer scale; PCS (Program Component Score) evaluates artistic dimensions on a quasi-continuous 0.25-increment scale. Normalized deviation is the leave-one-out score deviation divided by the era-specific item-level standard deviation (pre- or post-expansion, computed separately). The 2018/19 GOE expansion widened the scale from ±3 to ±5, nearly doubling the number of available grade levels.

4 Empirical Strategy

4.1 Baseline Specification

I estimate the same-nationality premium using:

$$y_{ijkc} = \beta \cdot \text{SameNation}_{jk} + \alpha_i + \gamma_j + \varepsilon_{ijkc} \quad (1)$$

where y_{ijkc} is the normalized score deviation of judge j on item i for skater k in competition c . Item fixed effects α_i , defined at the element-by-skater-segment level, absorb skater quality, competition identity, and segment difficulty in a single operator. Judge fixed effects γ_j absorb judge-specific leniency. Identification of β comes only from within-item variation in how different judges score the same performance. I cluster standard errors by judge; two-way clustering by judge-and-competition, judge-and-athlete-segment, or judge-and-skater leaves all standard errors within .001 of the single-way values.

Nine judges score the same skater on the same item, so β captures how much more a same-nationality judge deviates from the panel than different-nationality judges evaluating the identical performance.

4.2 Two Reforms as Joint Identification

Two ISU reforms in the past decade allow a joint test of the architectural mechanism. The 2018/19 GOE expansion changed the boundary architecture of one channel: the ISU widened the GOE range from ± 3 to ± 5 , increasing the number of integer levels from seven to eleven and narrowing the rule-defensibility region around each integer boundary, while leaving the 0.25-point PCS scale intact. The expansion was motivated by the technical goal of finer execution differentiation, not by concerns about evaluator bias ([International Skating Union, 2018](#)), so post-expansion declines in same-nationality bias reflect a behavioral response to architecture rather than purposeful enforcement. The 2022/23 PCS component reduction changed the holistic *content* of one channel

without touching its boundaries: it consolidated the five PCS components into three, but left the 0.25-point per-component grid intact (Appendix A.3). The architectural mechanism predicts that bias attenuates differentially after the architectural reform but not after the non-architectural reform.

A single saturated specification on the full panel disciplines this paired prediction:

$$\begin{aligned}
y_{ijkc} = & \beta_1 \cdot SN_{jk} \times GOE_i + \beta_2 \cdot SN_{jk} \times PCS_i \\
& + \beta_3 \cdot SN_{jk} \times GOE_i \times Post2018_c + \beta_4 \cdot SN_{jk} \times PCS_i \times Post2018_c \\
& + \beta_5 \cdot SN_{jk} \times GOE_i \times Post2022_c + \beta_6 \cdot SN_{jk} \times PCS_i \times Post2022_c \\
& + \alpha_i + \gamma_j + \varepsilon_{ijkc}
\end{aligned} \tag{2}$$

where $Post2018_c = 1$ for seasons after the 2018/19 reform and $Post2022_c = 1$ for seasons after the 2022/23 reform. PCS serves as the internal control for Reform A; GOE as the internal control for Reform B. Any season-level shock that shifts judging behavior symmetrically across domains is differenced out, so a nonzero triple difference reflects a domain-specific response to the corresponding reform.

Two triple differences encode the architectural predictions, each defined by placing the reform's affected channel first:

$$\Delta^A \equiv \beta_3 - \beta_4 \quad (\text{GOE attenuation relative to PCS, post-Reform A})$$

$$\Delta^B \equiv \beta_6 - \beta_5 \quad (\text{PCS attenuation relative to GOE, post-Reform B})$$

The architectural mechanism predicts $\Delta^A < 0$ and $\Delta^B \approx 0$. The joint signature is the F-test for $\Delta^A = \Delta^B$: rejecting equality, with Δ^A negative and Δ^B statistically small, is the empirical fingerprint of architectural cover: only reforms that change boundary architecture differentially attenuate compatriot bias.

Identifying assumption. Absent each reform, the same-nationality premium in the affected channel would have evolved in parallel with the premium in the unaffected channel, because the same

nine judges score the same skater simultaneously on both domains. Any unobserved shock that shifted judging behavior symmetrically across domains (a governance push, a cohort shift in the judge pool, a change in the skater pool) is absorbed by the within-performance counterfactual.

Defending the design. Threats to identification come in two flavors: those specific to Reform A and those specific to Reform B.

Reform A (2018/19, architectural). *Pre-trends:* the balanced-panel event study (Figure B1) restricts to judges and skaters present on both sides of 2018/19 and shows the GOE-minus-PCS differential flat through 2017/18 and breaking discretely thereafter; a pre-period placebo triple difference yields .008 (SE .007, $p = .24$), well below the post-expansion estimate, with 80%-power minimum detectable effect of .018 (Appendix B.4). *Anticipation:* ISU Communication 2168 announced the new GOE scale only in summer 2018, and the season-specific event study (Figure 1) shows no pre-expansion drift ([International Skating Union, 2018](#)). *Cross-domain spillover:* a within-judge carryover would attenuate Δ^A toward zero rather than create it, while substitution toward PCS would inflate the differential by depressing post-expansion GOE bias and elevating post-expansion PCS bias in opposite directions, a pattern inconsistent with the data since both channels' premiums fall after 2018/19 ($\beta_4 = -.016$, $p < .01$ in Table 3). *Co-timed reforms:* the only co-timed Sevilla-2018 decision plausibly relevant is Proposal 24's bias-screening expansion, whose baseline effect on GOE and PCS bias is symmetric (Section 5.4 discusses).

Reform B (2022/23, non-architectural). The 2022/23 component reduction was a discrete administrative change announced in spring 2022 with no co-timed architectural change to either domain. Within Equation (2), the post-2022 GOE coefficient ($\beta_5 = -.010$, $p < .01$) reflects continued post-2018 architectural dynamics rather than a 2022-specific shock; the post-2022 PCS coefficient is null ($\beta_6 = -.002$, not significant). Mechanical concerns about variance shifts from consolidating components are addressed in Section 5.3, where the boundary architecture relevant for notch upgrading—the 0.25-point per-component grid—was preserved. Compositional-stability and bandwidth checks for both reforms are summarized in Section 5.4 and Appendices B.1 and B.7.

DV scaling. For the continuous-DV specifications, the dependent variable divides each score deviation by the item-specific standard deviation computed separately in pre- and post-2018 eras. The 2018/19 expansion mechanically raised the item-level SD of raw GOE deviations by a factor of 1.304; PCS dispersion barely moved (ratio 0.945), and the 2022/23 PCS reduction generated no comparable shift. Table B7 reports five alternative scalings: the era-specific normalization (Col 1) and the non-ceiling-PCS subsample (Col 5) yield $\Delta^A = -.023$, weighing against ceiling compression as the source. Cols 2-4 (raw deviations, asinh, within-(item $\times$ season) SD) yield $\Delta^A > 0$ because PCS deviations span a wider raw range than GOE deviations; the constituent post-reform attenuation coefficients β_3, β_4 are negative in all five. Era-specific normalization, justified by the documented 2018 SD shift, is the appropriate common-scale DV for cross-domain comparison.

4.3 Notch Upgrading Tests

If bias operates through notch upgrading, same-nationality judges should be disproportionately likely to round up at discrete score boundaries (for analogous concentrations of responses at rule-defensible thresholds, see [Kleven and Waseem, 2013](#); [Dee et al., 2019](#); [Diamond and Persson, 2016](#)). I restrict the GOE sample to observations near the midpoint between two adjacent integers ($\pm.35$ window), retaining the central 70% of the between-integer region where individual judges could plausibly round either direction; Appendix B.7 confirms stability across bandwidths from $\pm.25$ to $\pm.45$. I estimate:

$$Upgrade_{ijkc} = \delta \cdot SN_{jk} + \theta \cdot SN_{jk} \times Post_c + \phi \cdot Distance_i + \mu_b + \gamma_j + \varepsilon_{ijkc} \quad (3)$$

where $Upgrade_{ijkc}$ equals one if the judge chose the higher adjacent integer, $Distance_i$ is the leave-one-out panel mean's distance to the boundary, μ_b are boundary fixed effects, and $Post_c$ equals one for seasons after the corresponding architectural reform: $Post_c = Post2018_c$ for the GOE channel, $Post_c = Post2022_c$ for the PCS upper-tail channel. The pooled premium δ identifies a same-nationality tendency to round up; the post-reform extension θ identifies the corresponding

reform’s effect on this premium, with the architectural prediction $\theta < 0$ for GOE (Reform A narrows the rule-defensibility region) and $\theta \approx 0$ for the PCS upper tail (Reform B leaves the boundary geometry intact).

For PCS, the regression (Table B4) is restricted to the three top-end midpoints $\{9.375, 9.625, 9.875\}$, where the perfect-10 ceiling makes the highest boundary the most rule-defensible point on the scale: any judge can defend giving an elite skater a 10.00. Figure 2 re-estimates Equation (3) separately at each boundary midpoint within each pre/post window for both reforms; Panel B’s PCS visualization spans the expanded upper-tail range $\{8.625, 8.875, 9.125, 9.375, 9.625, 9.875\}$ to display the boundary-by-boundary gradient.

5 Results

5.1 Compatriot Bias Is Larger on the Objective Channel

Same-nationality judges score compatriots more generously than other judges score the same skater on the same item: by .048 standard deviations of the item-level score-deviation distribution pooling both channels, by .060 on the objective GOE channel, and by .022 on the subjective PCS channel (Table 2, all $p < .01$). The GOE premium is nearly three times the PCS premium and equals roughly half of the typical disagreement among judges evaluating the identical performance (mean $|\text{normalized deviation}| = .127$). Aggregating across the items in an athlete-segment, a single same-nationality judge inflates a compatriot’s raw segment score by approximately .61 points after ISU panel-trimming.² The objective channel carries the larger premium, reversing the canonical pattern that bias concentrates on subjective scoring ([Zitzewitz, 2006](#)).

²The per-segment magnitude aggregates the per-item premium across items in an athlete-segment in original score units (raw same-nationality premium of 4.24 points; ISU rules drop the highest and lowest of nine judges’ marks per item, giving each judge effective weight $\approx 1/7$).

TABLE 2
BASELINE SAME-NATIONALITY PREMIUM

	(1) All	(2) GOE	(3) PCS
<i>Dep. var.: leave-one-out score deviation, normalized by era-specific item-level SD</i>			
Same nationality	.048*** (.002)	.060*** (.003)	.022*** (.001)
Item FE	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes
Mean abs(DV)	.097	.127	.032
Observations	1,012,339	693,885	318,454
Judge clusters	543	543	543
Within R-squared	.008	.009	.020

Note.—Unit of observation: judge $\times$ scoring item within an athlete-segment. Leave-one-out score deviation is the focal judge’s score minus the mean of the other eight judges on the same item; normalization divides by the era-specific item-level SD (pre- and post-2018/19 computed separately), so coefficients are in item-level SD units. All models include item and judge fixed effects. Standard errors in parentheses, clustered by judge.

*** Significant at the 1% level.

5.2 Only Architectural Reform Attenuates Bias

Two ISU reforms allow a joint test of the architectural mechanism. Only the 2018/19 GOE expansion changed boundary architecture by inserting four integer levels and narrowing the rule-defensibility region around each integer; the 2022/23 PCS component reduction left the 0.25-point per-component grid intact. The architectural mechanism predicts that only the architectural reform should differentially attenuate bias. The saturated specification estimated below interacts the same-nationality indicator separately with each domain (GOE, PCS), yielding pre-reform GOE and PCS premiums of .093 and .035; these differ from the unconditional Section 5.1 estimates (.060 and .022) because the saturated form nets out the cross-domain off-diagonal that an unconditional regression averages over.

The 2018/19 reform cut the GOE premium by 42%. From a pre-reform GOE premium of .093, the affected GOE channel falls by .039 ($p < .01$); the unaffected PCS channel falls by only .016 ($p < .01$). The architectural triple difference is $\Delta^A = -.023$ ($p < .001$, SE .004, Table 3 Column 1), the empirical signature of architectural attenuation. Figure 1 traces the parallel-trends evidence: GOE and PCS premiums move together in the two pre-expansion seasons, break discretely at 2018/19,

and the GOE–PCS differential persists through 2025/26.

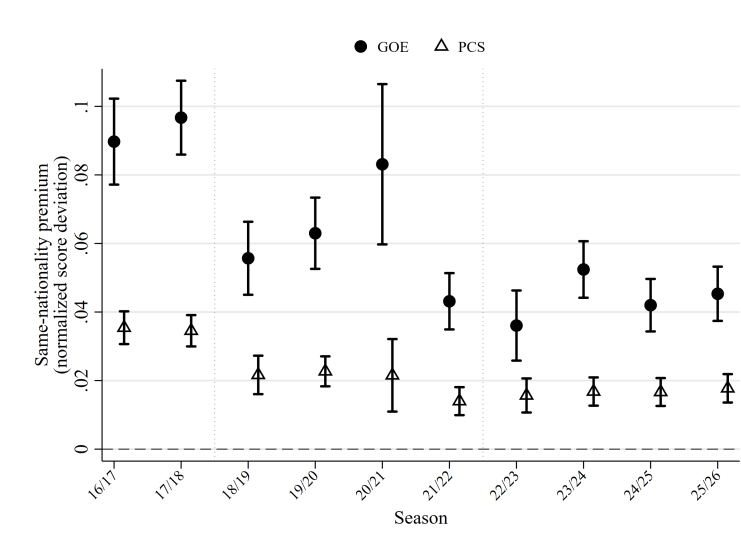


Figure 1: Same-Nationality Premiums by Season and Domain

Note.—Each point reports the season-specific same-nationality premium from a fully interacted regression of normalized score deviation on season $\times$ domain $\times$ same-nationality indicators, with item and judge fixed effects. Vertical bars show 95% confidence intervals (standard errors clustered by judge). Dotted lines mark the 2018/19 GOE expansion and 2022/23 PCS reduction.

The 2022/23 reform did not deliver comparable attenuation. The PCS premium is statistically unchanged ($-.002$, not significant) and the GOE premium continues a downward trend ($-.010$, $p < .01$) reflecting post-2018 dynamics rather than 2022-specific behavior. The non-architectural triple difference is $\Delta^B = +.008$ ($p < .05$). The sign is positive: PCS bias did not differentially attenuate post-Reform B, the opposite of what a generic-bias-decline alternative would predict. The redistribution that the architectural prediction implies is captured by the doubling at the 9.875 boundary (Table B4 Col 3 and Figure 2 Panel B) rather than by the average-level estimate.

The two reforms differ statistically. The F-test for $\Delta^A = \Delta^B$ rejects the null at $p < .001$ ($F = 43.3$, baseline column; $F \geq 24.1$ across all four robustness columns). Only architectural reform attenuates bias differentially.

TABLE 3
SATURATED TRIPLE-DIFFERENCE ACROSS TWO REFORMS

	(1) Baseline	(2) + Bloc control	(3) Balanced judges	(4) Judge $\times$ event FE
<i>Dep. var.: leave-one-out score deviation, normalized by era-specific item-level SD</i>				
<i>Pre-reform same-nationality bias</i>				
SN $\times$ GOE	.093*** (.005)	.094*** (.005)	.089*** (.005)	.095*** (.005)
SN $\times$ PCS	.035*** (.002)	.036*** (.002)	.033*** (.002)	.036*** (.002)
<i>Reform A: 2018/19 GOE Expansion (architectural; affected = GOE)</i>				
SN $\times$ GOE $\times$ Post-2018	-.039*** (.005)	-.039*** (.005)	-.034*** (.006)	-.039*** (.005)
SN $\times$ PCS $\times$ Post-2018	-.016*** (.002)	-.016*** (.002)	-.012*** (.002)	-.016*** (.003)
Δ^A (architectural)	-.023*** (.004)	-.023*** (.004)	-.022*** (.005)	-.023*** (.004)
<i>Reform B: 2022/23 PCS Reduction (non-architectural; affected = PCS)</i>				
SN $\times$ PCS $\times$ Post-2022	-.002 (.002)	-.002 (.002)	-.004* (.002)	-.003 (.002)
SN $\times$ GOE $\times$ Post-2022	-.010*** (.004)	-.010*** (.004)	-.010** (.005)	-.010*** (.004)
Δ^B (non-architectural)	.008** (.003)	.008** (.003)	.006 (.004)	.008** (.003)
Item FE	Yes	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes	—
Judge $\times$ event FE	—	—	—	Yes
<i>F-test (architectural = non-architectural)</i>	F = 43.3 p < .001	F = 43.3 p < .001	F = 24.1 p < .001	F = 44.0 p < .001
Mean abs(DV)	.097	.097	.098	.097
Observations	1,012,339	1,012,339	676,524	1,012,339
Judge clusters	543	543	219	543
Within R-squared	.010	.010	.010	.010

Note.—Unit of observation: judge $\times$ scoring item within an athlete-segment. *Post-2018* = 1 for seasons 2018/19 onward; *Post-2022* = 1 for seasons 2022/23 onward. All columns estimate a saturated triple-difference specification on the full panel with same-nationality interactions for both reforms. Δ^A is the lincom (SN $\times$ GOE $\times$ Post-2018) minus (SN $\times$ PCS $\times$ Post-2018), capturing the architectural 2018/19 GOE expansion. Δ^B is (SN $\times$ PCS $\times$ Post-2022) minus (SN $\times$ GOE $\times$ Post-2022), capturing the non-architectural 2022/23 PCS component reduction. The F-test reports equality of the two triple differences. Column (2) adds a same-continental-bloc control; Column (3) restricts to judges observed both before and after 2018/19; Column (4) replaces judge fixed effects with judge $\times$ event fixed effects. Standard errors in parentheses, clustered by judge.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

5.3 Notch Upgrading at Boundaries

Both reform results turn on what happens at boundaries between adjacent rule-coded scores. Tables 4 and B4 test whether notch upgrading—the same-nationality tendency to round up at boundaries—accounts for the cross-domain pattern.

5.3.1 GOE: Uniform Attenuation

At GOE integer-boundary midpoints, same-nationality judges round up 20.6 percentage points more often than panel peers pre-reform ($p < .01$, Table 4 Column 2), against a baseline upgrade rate of 50.8%. The 2018/19 expansion attenuates this premium to 14.7 percentage points post-reform, a 5.9 percentage point decline (29% reduction, $p < .01$). Panel A of Figure 2 shows the boundary-by-boundary pattern: same-nationality upgrade premiums are positive at every populated midpoint and the post-reform attenuation occurs across the scale, not at any single boundary, confirming that the reform operates through the architectural-boundary mechanism rather than through a uniform shift in the score distribution. The continuous-DV attenuation ($-.039$, Table 3) and the per-boundary attenuation (5.9 percentage points, Table 4) capture complementary channels: the conditional-on-near-boundary decline plus the mechanical shrinkage in the share of observations near wide-gap midpoints once the scale adds four new levels.

TABLE 4
GOE NOTCH UPGRADING

	(1) Pooled	(2) DD
<i>Dep. var.: 1 if judge chose the higher adjacent GOE integer notch, 0 otherwise</i>		
Same nationality	.161*** (.007)	.206*** (.011)
Same nationality $\times$ Post-GOE		-.059*** (.012)
Distance to boundary	.608*** (.006)	.608*** (.006)
Boundary FE	Yes	Yes
Judge FE	Yes	Yes
DV mean	.508	.508
Observations	409,594	409,594
Judge clusters	543	543
Within R-squared	.055	.055

Note.—Unit of observation: judge $\times$ scoring item within an athlete-segment. Sample restricted to GOE observations whose leave-one-out panel mean lies within ± 0.35 of the midpoint between two adjacent integers. Column (1) reports the pooled same-nationality upgrade premium across all integer boundaries. Column (2) adds the Same-nationality $\times$ Post-GOE interaction identifying the post-expansion change in the premium. Boundary-by-boundary visualization appears in Figure 2. All models include boundary and judge fixed effects. Standard errors in parentheses, clustered by judge.

*** Significant at the 1% level.

5.3.2 PCS: Redistribution to the Ceiling

The notch-upgrading mechanism extends to the top of the PCS scale, but with a different post-reform signature. Pooling across the three top-end midpoints (9.375, 9.625, 9.875), the same-nationality upgrade premium is 45.5 percentage points (Table B4 Col 1, $N = 13,015$). At the single boundary between 9.75 and a perfect 10.00, the premium is 59.0 percentage points (Col 3, $N = 984$, SE .093); the smaller sample yields a wider 95% confidence interval of [.41, .77] but the effect remains significant at $p < .01$. The 2022/23 reduction did not deliver uniform attenuation; instead, it *redistributed* bias up the scale. Panel B of Figure 2 shows the boundary-by-boundary pattern: the upgrade premium is modestly attenuated or flat at midpoints from 8.625 through 9.625, but at 9.875 it roughly doubles: from ≈ 56 to ≈ 115 percentage points, with non-overlapping confidence intervals. Bias did not dissipate; it concentrated at the maximally rule-defensible perfect-10 boundary, the only point on the PCS scale where any judge giving a 10.00 to an elite skater is fully defensible by the rulebook. Mid-scale placebos restricting attention away from the 10.00 ceiling yield

positive and significant same-nationality coefficients in every quartile of the midpoint distribution (Table B5); the effect is a property of the architectural boundary, not a ceiling artifact.

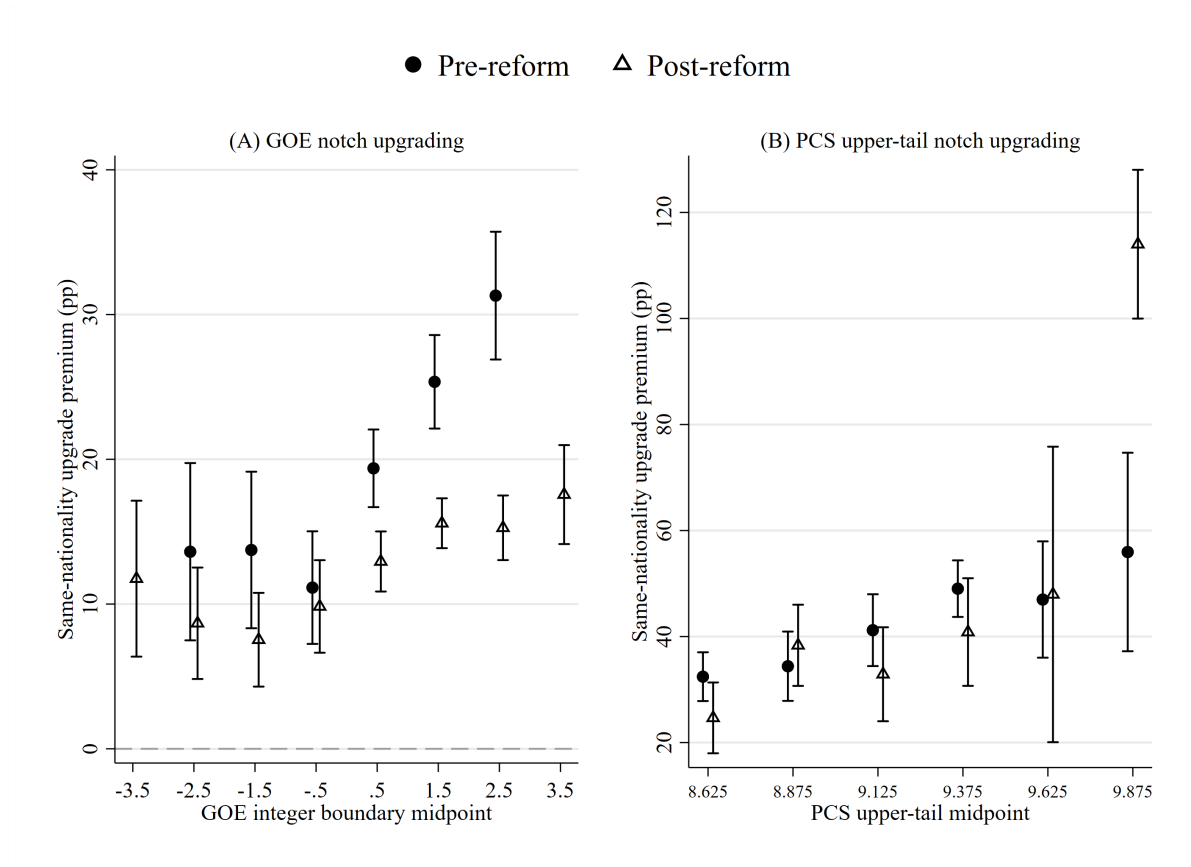


Figure 2: Notch Upgrading by Boundary, Pre vs. Post Reform

Note.—Each marker is the same-nationality coefficient from a separate regression at that boundary midpoint, with judge fixed effects and a distance-to-boundary control; vertical bars show 95% confidence intervals (standard errors clustered by judge). Panel A: GOE integer boundary midpoints over the symmetric window -3.5 to 3.5 , restricting observations to those whose leave-one-out panel mean lies within $\pm.35$ of the midpoint. Panel B: PCS upper-tail 0.25-grid midpoints from 8.625 to 9.875 , restricting observations to those whose panel mean lies within $\pm.10$ of the midpoint. The vertical scales differ between panels.

5.4 Robustness

The architectural triple difference $\Delta^A = -.023$ is preserved within $[-.029, -.019]$ across sample restrictions (balanced-judge, balanced-skater around 2018/19 and 2022/23, COVID-season exclusion, same-bloc controls; Table B1), FE specifications (judge $\times$ event; Table 3 Col 4: $\Delta^A = -.023$), and the non-ceiling-PCS subsample (Table B7 Col 5: $\Delta^A = -.023$). The notch-upgrading speci-

fications are stable across boundary-window widths from $\pm.25$ to $\pm.45$ for GOE and $\pm.05$ to $\pm.15$ for PCS (Table B6). DV-scaling sensitivity is discussed in Section 4.2. Pre-trends are flat: the balanced-panel event study (Figure B1) shows the GOE–PCS differential level through 2017/18 and breaking discretely at 2018/19, and a pre-period placebo triple difference yields .008 ($p = .24$, Appendix B.4). Notch upgrading is driven by national identity rather than shared cultural scoring norms: replacing same-nationality with same-continental-bloc yields near-zero upgrading effects (Table B2). Away from boundaries, the GOE premium (.033) is close to the PCS premium (.022), consistent with boundary cases accounting for the bulk of the cross-domain gap (Table B3).

Co-timed governance shocks. The 2018/19 Seville Congress bundled multiple judging-related decisions; the only one plausibly relevant for the triple difference is Proposal 24, which expanded the Officials Assessment Commission’s mandate to screen for suspected national bias. Because the OAC screens deviations from panel consensus on every scored item, a domain-invariant detection apparatus, its baseline effect on GOE and PCS bias is symmetric, leaving the architectural triple difference attributable to the scale expansion.

6 Discussion

6.1 Hiding in Plain Sight

What earlier audits identified as the locus of compatriot bias in figure skating, the more subjective scoring channel ([Zitzewitz, 2006](#)), is no longer where bias lives. Over twenty years, the subjective margin has been the target of reforms, of a more structured scoring system, and of sustained media scrutiny. Single-channel audits cannot distinguish bias eliminated from bias relocated, and public measurement systems often induce strategic adjustment rather than genuine change ([Espeland and Sauder, 2007](#)). The framing of evaluation as objective or meritocratic can paradoxically license bias rather than reduce it: managers told their organization rewards performance meritocratically give smaller bonuses to women than to men with identical performance ([Castilla and Benard, 2010](#)), the opposite of the framing’s intent. Debiasing information updates loan officers’ beliefs but leaves their

taste-based discrimination intact ([Montoya et al., 2026](#)), illustrating that interventions miss bias when they target the wrong locus. Figure skating's two-channel architecture reveals the geography. The same nine judges scoring the same performance simultaneously on an objective GOE channel and a subjective PCS channel deposit a same-nationality premium nearly three times larger on the objective channel. The residual bias is concentrated on the channel the reforms certified as objective: not where bias is absent, but where it is least visible. The audit logic treats rule-coded scoring as transparent, and the judges themselves can record the higher of two rule-defensible scores without necessarily recognizing the choice as biased.

Each integer boundary on the GOE scale resembles an unmarked balance: two adjacent scores are simultaneously rule-defensible, and the rule book offers no instrument for choosing between them. The mechanism is rule-defensibility, not subjectivity. Discrete thresholds of this kind also pull on the agent side: athletes and exam-takers exert extra effort to cross round numbers ([Pope and Simonsohn, 2011](#)), and charities adjust their reported finances to bunch just above public-rating star cutoffs ([Mayo, 2025](#)). Here the architecture transfers that pull to the evaluator, and bias enters as the systematic tilt across thousands of such balances. The notch-upgrading test makes the tilt visible: at integer-boundary midpoints, same-nationality judges record the higher of two adjacent scores 20.6 percentage points more often than panel peers. The same mechanism produces a 45.5-point premium across PCS's top-end midpoints, the region surrounding its one architectural boundary at the perfect-10 ceiling. Two reforms then reveal the architectural geography in mirror image. The 2018/19 reform attenuated the bias by inserting four integer levels and shrinking the rule-defensibility region around each GOE boundary. The 2022/23 reform held boundary architecture fixed and consolidated five PCS components into three; bias did not dissipate but migrated, roughly doubling at the perfect-10 ceiling, the maximally rule-defensible boundary on the PCS scale. Where rules permit two defensible scores bias settles; where a reform leaves a defensible boundary standing, bias finds it.

The geography arises from the joint operation of two factors: a motivated agent and an architecture that authorizes the preferred score as rule-compliant. The first factor is the psychological

substrate established in the motivated reasoning literature: preferred conclusions can be reached only when actors construct seemingly reasonable justifications (Kunda, 1990), and ambiguity supplies the moral wiggle room such reasoning needs to operate (Dana et al., 2007). Zitzewitz (2006, footnote 13) suggested two decades ago that nationalistic favoritism in figure skating may operate without judges being fully conscious of it. The second factor is the architecture that supplies the justification: biased acts that cannot be objectively identified as discriminatory persist through plausible deniability (Pikulina and Ferreira, 2026), and in the psychological parallel, latent biases are expressed when norms suppressing them are released by acceptable justifications (Crandall and Eshleman, 2003). The rule book’s certification of two adjacent scores as defensible is the institutional form of such a justification. Parsons et al. (2011) provide the closest field analog at a single architectural point: in baseball umpiring, in-group bias is amplified at edge-zone pitches where the rules give umpires maximum discretion. What this paper adds is system-wide: the architectural geography across a full scoring system, and how that geography responds when one channel of the architecture is reformed. The 2018/19 reform’s roughly 42% attenuation of the GOE compatriot premium is what the joint-product framing predicts: a reform shrinks the rule-defensibility surface but does not remove the motivated reasoning that operates on what remains.

6.2 Implications for Scale Design

The relationship between scale granularity and bias is non-monotonic, depending on which mechanism is active. Rivera and Tilcsik (2019) show that narrowing a teaching-evaluation scale from ten to six points reduces gender bias because the rating “10” evokes brilliance stereotypes that disadvantage women, while “6” does not. Botelho et al. (2025) show that dichotomizing a continuous online-platform rating eliminates the racial penalty by removing the intermediate scale points where subtle downgrading operates.

The findings here add a third pattern: when bias concentrates at architectural boundaries through notch upgrading, inserting additional integer levels narrows the rule-defensibility region around each boundary and attenuates the bias.

There is no universal prescription for more or fewer scale points; the appropriate redesign depends on diagnosis. Where notch upgrading dominates, finer granularity attenuates bias by shrinking architectural defensibility regions. Where extreme scale values activate stereotypes, narrowing the scale to remove those values may help. Where bias takes the form of subtle downgrades at intermediate scale points, dichotomization may help. The first design step is identifying which mechanism is active.

6.3 Generalizing Beyond Figure Skating

The framework provides a diagnostic toolkit rather than a finished claim about other settings. Three conditions are necessary for the notch-upgrading mechanism to operate: a discrete-level rating architecture with rule-coded boundaries; evaluators with directional preferences relative to the units being rated; and sufficient observability of boundary-midpoint behavior to detect asymmetric upgrading. Where these conditions hold, the empirical specifications used here transfer. Where they do not, including continuous scoring, single-channel evaluations without within-unit comparison, or settings where the rules do not codify level distinctions, the mechanism does not apply. Other interventions operate on different ingredients of bias: removing demographic visibility ([Goldin and Rouse, 2000](#)), shifting from separate to joint evaluation ([Bohnet et al., 2016](#)), or replacing human prediction with algorithms ([Kleinberg et al., 2018](#)). Categorical performance ratings and credit-tier letter grades share the surface architectural feature, but empirical confirmation in those settings, including verification of rule-coded boundary structure and direct observation of boundary-midpoint behavior, remains future work. The within-channel mechanism studied here is distinct from cross-stage systemic discrimination, where direct discrimination at one decision node compounds through subsequent stages ([Bohren et al., 2025b](#)): that work explains how bias propagates across nodes; this paper explains where bias hides within a single node.

The external-validity profile of the present findings follows the SANS framework ([List, 2020](#)). *Selection*: judges are elite ISU-licensed international officials, a high-quality sample whose estimates of compatriot favoritism likely understate the effect in less-monitored settings. *Attrition*:

the balanced-panel event study (Figure B1) fixes judge composition across the 2018/19 reform. *Naturalness*: ISU competitions are high-stakes natural evaluations with careers, prize money, and Olympic qualification at stake, not laboratory analogs. *Scaling*: findings document the architectural mechanism within the senior international ISU circuit; transfer to other rule-coded evaluation systems requires verification of the architectural conditions outlined above.

6.4 Limitations

Three limitations warrant note. First, figure skating is a specific context, but the architectural mechanism is not specific to nationality: it operates wherever discrete boundaries create rule-defensibility regions for evaluators with directional preferences. Second, GOE and PCS differ along dimensions beyond boundary width; the within-performance triple-difference design addresses third-factor confounds but cannot fully isolate boundary width from every other architectural feature distinguishing the two channels. Third, I observe scoring behavior but not judge intentions and cannot adjudicate between unconscious bias and deliberate strategy (cf. [Barron et al., 2025](#)), nor among the underlying sources of discrimination—taste-based animus, accurate statistical inference, or inaccurate beliefs about group quality ([Bohren et al., 2025a](#)); the architectural mechanism is consistent with any of these, since architectural cover supplies what any form of motivated choice requires.

7 Conclusion

Audits of structured evaluation systems should test for asymmetric behavior at architectural boundaries, not only for bias on subjective margins. The bias that audits no longer find on subjective channels has not been eliminated; it persists where the rules certify two adjacent scores as defensible. In figure skating, compatriot bias on the more objective GOE channel is nearly three times larger than on the more subjective PCS channel, concentrated at the integer-boundary midpoints where the rule book offers no instrument for choosing between adjacent scores.

The architecture of evaluation is not neutral. Discrete rule-coded boundaries create regions of

rule-defensibility, and motivated reasoning settles where those regions exist. The 2018/19 ISU reform attenuated GOE compatriot bias by roughly 42%, not by reducing the subjectivity of the channel but by inserting four additional integer levels and shrinking the rule-defensibility region around each boundary. The same mechanism produced a 45.5-point notch-upgrading premium across PCS's top-end midpoints, the region around its perfect-10 ceiling. A complementary 2022/23 PCS reform that consolidated components without altering boundary architecture produced no comparable attenuation; instead, post-reform bias roughly doubled at the perfect-10 boundary that remained. Architectural reform can shrink bias or relocate it, but the architecture itself, not the channel's subjectivity, determines which.

Reforms that shrink the rule-defensibility region attenuate bias; reforms that consolidate scale content while leaving architectural cover intact do not. The bias hides where the audit logic does not look.

Data Availability

Replication code and the analytic dataset of 1,012,339 judge-item observations constructed for this study will be deposited in the Harvard Dataverse upon acceptance. The underlying primary source—ISU protocol sheets including individual judge identities and scores from 2016/17 onward—is publicly available at <https://www.isu.org>.

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Appendix A Institutional Background

A.1 The ISU Judging System

Since the 2004/05 season, the International Skating Union (ISU) has scored figure skating competitions using the ISU Judging System (IJS). A panel of nine judges independently evaluates each performance along two dimensions. The *Grade of Execution* (GOE) measures the technical quality of individual elements (jumps, spins, step sequences). Each element carries a base value determined by the ISU technical committee; each judge adjusts this value by assigning an integer grade. The *Program Component Score* (PCS) captures broader skating qualities; judges rate each component from .25 to 10.00 in 0.25-point increments. For each scored item, the highest and lowest of the nine marks are trimmed and the remaining seven are averaged. Individual judges' marks have been publicly identified on ISU protocol sheets since the 2016/17 season, the first season of the sample used in this paper.

A.2 The 2018/19 GOE Scale Expansion

At the 57th Ordinary Congress (Seville, June 4–8, 2018), the ISU voted to expand the GOE scale from ± 3 (seven integer levels) to ± 5 (eleven integer levels), effective from the 2018/19 season. The implementing regulations were codified in Communication No. 2168, dated May 23, 2018, and later updated by Communication No. 2186 ([International Skating Union, 2018](#)). Under the new scale, each GOE level corresponds to a 10% increment of the element's base value (e.g., $+5 = +50\%$, $-3 = -30\%$). The stated rationale was to permit finer differentiation of execution quality. The expansion changed the boundary structure of GOE by narrowing the distance between adjacent integer levels, without altering any aspect of PCS scoring.

The 2018/19 Seville Congress bundled multiple judging-related decisions, but the GOE scale expansion is the only one that changed the boundary structure of one scoring domain while leaving the other unchanged. Among the co-timed decisions, only Proposal 24, which expanded the Officials Assessment Commission's mandate to screen for suspected national bias, is plausibly rel-

evant. Because Proposal 24 is a governance shock that should reduce bias in *both* domains equally, it cannot explain the differential decline in GOE bias relative to PCS bias that the triple-difference design identifies. No contemporaneous reform altered the PCS boundary structure.

A.3 The 2022/23 PCS Component Reduction

At the 58th Ordinary Congress (Phuket, June 6–10, 2022), the ISU adopted Proposals 158 and 201, which consolidated the five PCS categories into three: Skating Skills, Composition, and Presentation ([International Skating Union, 2022](#)). The previous categories of Transitions and Interpretation were absorbed into the new categories. The change took effect in the 2022/23 season. Crucially, the 0.25-point scoring scale and the .25–10.00 range were unchanged; only the number of scored components was reduced. PCS factors were adjusted so that the total PCS contribution to segment scores remained approximately equivalent. This reduction altered the number of PCS items without changing the boundary structure of individual PCS scores.

Appendix B Supplementary Results

B.1 Robustness

Table B1 tests the sensitivity of the saturated triple-difference specification across sample subsets. Column (1) reproduces the main specification. Column (2) restricts the sample to 219 judges who appear both before and after the 2018/19 expansion; the architectural triple difference Δ^A is close to the main-spec value, indicating the post-expansion attenuation is not driven by judge composition changes. Column (3) restricts the sample to skaters observed both before and after the 2018/19 expansion, holding the competitor pool fixed across Reform A rather than the evaluator pool; Δ^A remains close to the main-spec value. Column (4) adds an indicator for same-continental-bloc pairs. Bloc favoritism is small and does not change the same-nationality estimates. Column (5) drops the 2019/20 COVID-affected season; results are again unchanged. Column (6) restricts the sample to skaters observed both before and after the 2022/23 PCS reduction, closing the symmetry between the Reform A and Reform B robustness checks; the architectural attenuation strengthens slightly to $\Delta^A = -.029$ and the non-architectural triple difference remains small at $\Delta^B = +.011$. In all six columns, Δ^A is negative and significant while Δ^B is small and economically modest, confirming the asymmetric architectural prediction.

TABLE B1
SATURATED TRIPLE-DIFFERENCE: ROBUSTNESS

	(1)	(2)	(3)	(4)	(5)	(6)
	Main spec	Balanced judges	Balanced skaters (2018)	+ Bloc control	No COVID	Balanced skaters (2022)
<i>Dep. var.: leave-one-out score deviation, normalized by era-specific item-level SD</i>						
SN × GOE	.093*** (.005)	.089*** (.005)	.088*** (.005)	.094*** (.005)	.093*** (.005)	.097*** (.011)
SN × PCS	.035*** (.002)	.033*** (.002)	.034*** (.003)	.036*** (.002)	.035*** (.002)	.032*** (.006)
SN × GOE × Post-2018	-.039*** (.005)	-.034*** (.006)	-.032*** (.006)	-.039*** (.005)	-.042*** (.005)	-.046*** (.011)
SN × PCS × Post-2018	-.016*** (.002)	-.012*** (.002)	-.013*** (.003)	-.016*** (.002)	-.017*** (.002)	-.017*** (.006)
SN × GOE × Post-2022	-.010*** (.004)	-.010** (.005)	-.005 (.006)	-.010*** (.004)	-.006 (.004)	-.008* (.004)
SN × PCS × Post-2022	-.002 (.002)	-.004* (.002)	-.000 (.003)	-.002 (.002)	-.001 (.002)	.004 (.003)
Same bloc, different nationality				.008*** (.001)		
Item FE	Yes	Yes	Yes	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes	Yes	Yes	Yes
Δ^A (architectural)	-.023	-.022	-.019	-.023	-.025	-.029
Δ^B (non-architectural)	.008	.006	.005	.008	.005	.011
Mean abs(DV)	.097	.098	.098	.097	.098	.094
Observations	1,012,339	676,524	405,954	1,012,339	921,178	411,899
Judge clusters	543	219	539	543	537	539
Within R-squared	.010	.010	.011	.010	.010	.008

Note.—All columns estimate the saturated triple-difference specification on the corresponding sample subset; Δ^A and Δ^B are the architectural and non-architectural triple differences (defined in Table 3). Column (2) restricts to judges observed both before and after 2018/19; Column (3) restricts to skaters observed both before and after 2018/19; Column (4) adds a same-continental-bloc control; Column (5) excludes the 2019/20 COVID-affected season; Column (6) restricts to skaters observed both before and after 2022/23. All models include item and judge fixed effects. Standard errors in parentheses, clustered by judge.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

B.2 Correlated Rounding

A potential concern is that notch upgrading reflects correlated rounding: judges and skaters from the same cultural background may resolve ambiguous cases similarly, even absent favoritism. Table B2 addresses this by replacing the same-nationality indicator with a same-continental-bloc indicator (excluding same-country pairs). Same-bloc upgrading at GOE boundaries is small and at most marginally significant (.007, $p < .10$, Column (1)) and vanishes in the pre/post split (Column (2)). When both indicators enter simultaneously, same-nationality upgrading remains large (.166–.210, $p < .01$) and declines after the expansion ($-.058$, $p < .01$), while the same-bloc coefficient is small and does not respond to the expansion (Columns (3)–(4)). The same pattern holds at PCS top-end boundaries (Columns (5)–(6)). Notch upgrading is driven by national identity, not shared cultural scoring norms.

TABLE B2
SAME-BLOC NOTCH UPGRADING (PLACEBO)

	GOE boundary		GOE boundary (horse race)		PCS top-end	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Dep. var.: 1 if judge chose the higher adjacent score value, 0 otherwise</i>						
Same nationality			.166*** (.007)	.210*** (.011)		.454*** (.025)
Same bloc, diff. nationality	.007* (.004)	-.000 (.009)	.023*** (.004)	.017* (.009)	-.048** (.024)	-.005 (.019)
Same nationality $\times$ Post-GOE				-.058*** (.012)		
Same bloc $\times$ Post-GOE		.010 (.010)		.008 (.010)		
Distance to boundary	.608*** (.006)	.608*** (.006)	.608*** (.006)	.608*** (.006)	1.865*** (.075)	1.865*** (.074)
Boundary FE	Yes	Yes	Yes	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes	Yes	Yes	Yes
DV mean	.508	.508	.508	.508	.488	.488
Observations	409,594	409,594	409,594	409,594	13,015	13,015
Judge clusters	543	543	543	543	459	459
Within R-squared	.047	.047	.055	.055	.059	.105

Note.—Columns (1)–(2) replace same nationality with same continental bloc (different country) in the GOE boundary sample. Columns (3)–(4) include both indicators simultaneously. Columns (5)–(6) repeat the exercise for PCS top-end boundaries. Same-bloc pairs share a continental group (East Asia, North America, Western Europe, or Eastern Europe) but differ in nationality. All models include boundary and judge fixed effects. Standard errors in parentheses, clustered by judge.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

B.3 Mechanism Accounting

Table B3 quantifies how much of the GOE–PCS bias gap is accounted for by boundary cases. GOE observations are classified as near-boundary (leave-one-out panel mean within $\pm.35$ of the midpoint between adjacent integers) or non-boundary. Column (1) shows overall GOE bias of .060 standard deviations. Splitting by boundary proximity, near-boundary observations show bias of .049 (Column (2)), while non-boundary observations show .033 (Column (3)). PCS bias is .022 (Column (4)). Non-boundary GOE bias (.033) is much closer to PCS bias (.022) than to overall GOE bias (.060), suggesting that boundary cases account for most of the GOE–PCS gap. Columns (5)–(6) add the pre/post interaction: same-nationality bias declines after the expansion in both near-boundary ($-.044$, $p < .01$) and non-boundary ($-.030$, $p < .01$) GOE observations.

TABLE B3
BOUNDARY VS. NON-BOUNDARY GOE BIAS

	(1)	(2)	(3)	(4)	(5)	(6)
		<i>Pre+post pooled</i>			<i>With Post-GOE interaction</i>	
	All GOE	Near-bdry	Non-bdry	PCS	Near-bdry	Non-bdry
<i>Dep. var.: leave-one-out score deviation, normalized by era-specific item-level SD</i>						
Same nationality	.060*** (.003)	.049*** (.002)	.033*** (.002)	.022*** (.001)	.082*** (.005)	.055*** (.004)
Same nationality $\times$ Post-GOE					-.044*** (.005)	-.030*** (.004)
Item FE	Yes	Yes	Yes	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean abs(DV)	.127	.139	.101	.032	.139	.101
Observations	693,885	401,812	280,489	318,454	401,812	280,489
Judge clusters	543	543	543	543	543	543
Within R-squared	.009	.007	.005	.020	.008	.006

Note.—GOE observations are classified as near-boundary (leave-one-out panel mean within ± 0.35 of the midpoint between adjacent integers) or non-boundary. Columns (1)–(4) pool pre- and post-expansion observations and report the overall same-nationality premium. Columns (5)–(6) additionally interact same-nationality with the post-expansion indicator: the first coefficient row reports the pre-expansion same-nationality effect and the second row reports the post-expansion change, with the *With Post-GOE interaction* spanner in the column header flagging this specification. All models include item and judge fixed effects. Standard errors in parentheses, clustered by judge.

*** Significant at the 1% level.

B.4 Pre-Period Placebo, Event Study, and Minimum Detectable Effects

A two-season pre-expansion window limits the conventional F -test for parallel trends. This subsection reports a pre-period placebo triple difference and a balanced-panel event study, with minimum detectable effect (MDE) calculations to bound the implied power.

Pre-period placebo. Restricting the sample to 2016/17 and 2017/18 and treating 2017/18 as a pseudo-post-expansion season, the triple-difference specification yields a GOE-minus-PCS differential of .008 (SE .007, $p = .24$): a clean null on observations uncontaminated by the 2018/19 reform. At 80% power, the architectural triple difference's MDE is .011 (48% of the observed $-.023$), and the pre-period placebo's MDE is .018, well above the observed pre-period differential. The non-architectural triple difference $\Delta^B = +.008$ has an 80%-power MDE of .008, so the average-level Reform B null sits at the power threshold; the redistribution evidence in Section 5.3.2 (Table B4 Col 3) does the load-bearing work. The null findings are genuine absences at economically meaningful thresholds, not power-starved non-rejections.

Balanced-panel event study. Figure B1 estimates season-specific same-nationality premiums within the balanced panel of 219 judges observed both before and after the expansion, fixing judge composition across the reform. In the two pre-expansion seasons, the GOE premium is statistically flat and the 95% confidence intervals include the baseline level; the premium drops sharply in 2018/19 and settles at a lower level through 2025/26. The PCS premium moves only modestly throughout. The 80%-power minimum detectable drift across the two pre-expansion seasons is $\approx .010$, well above the observed pre-period coefficient differences.

B.5 PCS Top-End Notch Upgrading

Table B4 reports the notch-upgrading test at the top of the PCS scale. I restrict the sample to 13,015 PCS observations whose leave-one-out panel mean lies within $\pm .10$ of one of three top-end midpoints: 9.375, 9.625, or 9.875. The dependent variable equals one if the judge chose the higher

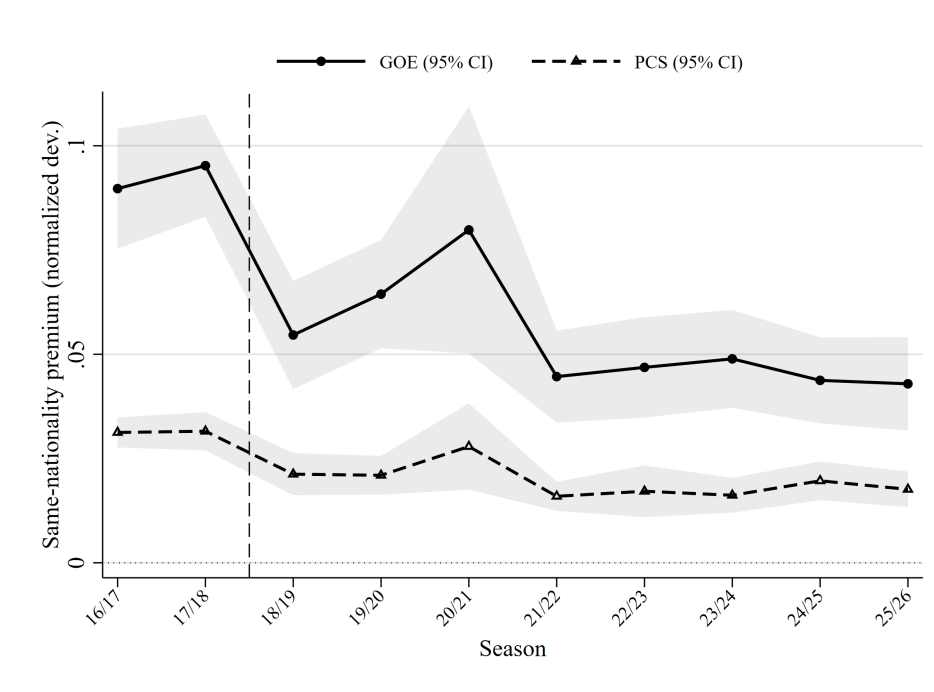


Figure B1: Balanced-Panel Event Study of Same-Nationality Premium

Note.—Sample restricted to the 219 judges observed in at least one pre-expansion season (2016/17–2017/18) and at least one post-expansion season (2018/19 onward), fixing judge composition across the reform. Each point plots the season-specific same-nationality premium from a fully interacted regression of normalized score deviation on season $\times$ domain $\times$ same-nationality indicators, with item and judge fixed effects. Vertical bars show 95% confidence intervals (standard errors clustered by judge). The dotted vertical line marks the 2018/19 GOE expansion. The GOE premium is statistically flat in the two pre-expansion seasons and drops sharply in 2018/19; the PCS premium moves only modestly throughout the sample.

adjacent .25 grid value. Column (1) shows that same-nationality judges are 45.5 percentage points more likely to choose the higher option ($p < .01$), roughly 2.8 times the GOE same-nationality upgrade premium documented in Table 4. Column (2) separates the pre- and post-2022/23 periods: the gap is 46.9 points before the PCS reduction and falls by 7.6 points afterward ($p < .10$; the post-2022/23 window spans three seasons and the top-end PCS boundary sample is roughly a thirtieth the size of the GOE boundary sample, so power for detecting a decline is correspondingly lower than on the GOE side).

Column (3) isolates a narrower subsample restricted to the top-most midpoint, 9.875, where the judge's choice is between 9.75 and a perfect 10.00; this single-midpoint estimate serves as a supplementary diagnostic on the 10.00 ceiling and is not directly comparable to the pooled Columns (1)–(2). Within this subsample, same-nationality judges are 59.0 percentage points more likely to award a perfect 10.00 ($p < .01$). A ceiling-specific interpretation is addressed by the mid-scale placebo in Table B5, which finds positive and highly significant same-nationality coefficients in every quartile of the midpoint distribution.

TABLE B4
PCS TOP-END NOTCH UPGRADING

	(1) Pooled	(2) DD	(3) Perfect 10
<i>Dep. var.: 1 if judge chose the higher adjacent 0.25-grid PCS value, 0 otherwise</i>			
Same nationality	.455*** (.025)	.469*** (.027)	.590*** (.093)
Same nationality $\times$ Post-PCS		-.076* (.045)	
Distance to boundary	1.865*** (.074)	1.865*** (.074)	2.803*** (.238)
Boundary FE	Yes	Yes	No
Judge FE	Yes	Yes	Yes
DV mean	.488	.488	.458
Observations	13,015	13,015	984
Judge clusters	459	459	173
Within R-squared	.105	.106	.190

Note.—Sample restricted to PCS observations near top-end midpoints {9.375, 9.625, 9.875} with window ± 0.10 around each midpoint. *Post-PCS* equals one for competition seasons 2022/23 and later (distinct from the 2018/19 GOE expansion). Column (3) isolates the top-most midpoint 9.875 (the choice between 9.75 and a perfect 10.00). Columns (1)–(2) include boundary and judge fixed effects; Column (3) includes judge fixed effects only. Standard errors in parentheses, clustered by judge. See Table B5 for the full-scale version (Panel B) that rules out a ceiling-specific interpretation.

* Significant at the 10% level.

*** Significant at the 1% level.

B.6 Full-Scale Notch Upgrading: Mid-Scale Placebos

The notch-upgrading tests in the main text (Table 4) and Table B4 focus on score boundaries where the effect is most readily detectable: all populated GOE integer midpoints and the three top-end PCS midpoints {9.375, 9.625, 9.875}. A natural concern is that the bias might operate exclusively at scale extremes (the ± 4.5 GOE midpoints with thin support or the PCS 10.00 ceiling). Table B5 performs the mid-scale placebo for both scales in a single 2-panel layout, estimating the same-nationality effect on every populated midpoint and splitting the sample by empirical quartile of boundary location.

TABLE B5
MID-SCALE PLACEBO TESTS

	Full scale		By empirical quartile of boundary			
	(1) Pooled	(2) DD	(3) Q1 (low)	(4) Q2	(5) Q3	(6) Q4 (high)
Panel A: GOE scale (<i>DV: 1 if higher adjacent integer notch</i>)						
Same nationality	.161*** (.007)	.206*** (.011)	.130*** (.007)	.184*** (.008)	.173*** (.011)	.180*** (.017)
Same nationality × Post-GOE		-.059*** (.012)				
Distance to boundary	.608*** (.006)	.608*** (.006)	.614*** (.008)	.567*** (.009)	.617*** (.010)	.672*** (.015)
Observations	409,594	409,594	152,176	124,070	94,715	38,630
Panel B: PCS scale (<i>DV: 1 if higher adjacent 0.25-grid value</i>)						
Same nationality	.245*** (.010)	.263*** (.011)	.193*** (.014)	.226*** (.011)	.274*** (.014)	.349*** (.017)
Same nationality × Post-PCS		-.055*** (.014)				
Distance to boundary	1.077*** (.018)	1.077*** (.018)	.857*** (.029)	.992*** (.029)	1.104*** (.033)	1.469*** (.037)
Observations	278,173	278,173	70,251	93,361	63,651	50,896
Boundary FE	Yes	Yes	Yes	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes	Yes	Yes	Yes

Note.—Combined mid-scale placebo for the notch-upgrading mechanism. Each panel reports Cols (1) pooled across all populated midpoints; Col (2) adds the pre/post-reform interaction (Post-GOE in Panel A, Post-PCS in Panel B); Cols (3)–(6) split the sample by empirical quartile of the observation-weighted boundary distribution. Panel A quartile cutoffs (GOE boundary value): $Q1 \leq 0.50$, $Q2 \leq 1.50$, $Q3 \leq 2.50$, $Q4 \geq 3.50$. Panel B quartile cutoffs (PCS midpoint): $Q1 \leq 6.625$, $Q2 \leq 7.625$, $Q3 \leq 8.375$, $Q4 \geq 8.625$. Positive same-nationality coefficients across all four quartiles in both panels rule out scale-extremes (Panel A) or 10.00-ceiling (Panel B) interpretations: the behavior operates throughout each scale. All models include boundary and judge fixed effects; standard errors in parentheses, clustered by judge.

*** Significant at the 1% level.

For GOE (Panel A), the pooled effect is .161 ($p < .01$). The quartile split shows positive and highly significant coefficients throughout (.130, .184, .173, .180), with the smallest effect at the low end (Q1, boundaries below .50) and the largest at the middle of the populated range (Q2–Q4). The notch-upgrading behavior is not concentrated at the ± 4.5 GOE scale extremes; it operates throughout the scale. For PCS (Panel B), the pooled same-nationality coefficient is .245 ($p < .01$) across all midpoints from .125 to 9.875. Splitting by empirical quartile produces a monotone gradient: the effect rises from .193 in Q1 to .349 in Q4. All four quartiles are positive and highly significant; the low-midpoint Q1 effect rules out the interpretation that the baseline top-end PCS result is driven solely by the 10.00 ceiling.

B.7 Boundary-Window Width Sensitivity

The baseline notch-upgrading specifications define near-boundary observations using a fixed window around each midpoint: $\pm .35$ for GOE and $\pm .10$ for PCS. Table B6 re-estimates the two main notch-upgrading regressions across three window widths. For GOE (Columns 1–3), same-nationality upgrading is .161 at $\pm .25$, .161 at $\pm .35$ (baseline), and .154 at $\pm .45$; the effect is insensitive to window choice within the plausible range. For PCS (Columns 4–6), the narrower window of $\pm .05$ contains only 5,515 observations around three midpoints and yields a .442 effect; widening to $\pm .10$ (baseline) and $\pm .15$ produces .455 and .454 on progressively larger samples, also stable. The baseline windows are neither cherry-picked nor an artifact of bandwidth choice.

TABLE B6
BOUNDARY-WINDOW WIDTH SENSITIVITY

	GOE boundary			PCS top-end boundary		
	(1) ±0.25	(2) ±0.35	(3) ±0.45	(4) ±0.05	(5) ±0.10	(6) ±0.15
<i>Dep. var.: 1 if judge chose the higher adjacent scale value, 0 otherwise</i>						
Same nationality	.161*** (.007)	.161*** (.007)	.154*** (.006)	.442*** (.038)	.455*** (.025)	.454*** (.024)
Distance to boundary	.608*** (.006)	.608*** (.006)	.598*** (.005)	1.992*** (.246)	1.865*** (.074)	1.739*** (.064)
Boundary FE	Yes	Yes	Yes	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes	Yes	Yes	Yes
DV mean	.508	.508	.505	.489	.488	.505
Observations	409,389	409,594	581,782	5,515	13,015	14,276
Judge clusters	543	543	543	413	459	460
Within R-squared	.054	.055	.098	.057	.105	.111

Note.—Cols (1)–(3) estimate the GOE boundary-pushing specification (Table 4) at window widths of ±0.25, ±0.35 (baseline), and ±0.45 around each half-integer midpoint. Cols (4)–(6) estimate the PCS top-end boundary-pushing specification (Table B4) at window widths of ±0.05, ±0.10 (baseline), and ±0.15 around the three top-end midpoints {9.375, 9.625, 9.875}. The GOE dependent variable is an indicator for the higher adjacent integer notch; the PCS dependent variable is an indicator for the higher adjacent 0.25 grid value. All models include boundary and judge fixed effects. Standard errors in parentheses, clustered by judge.

*** Significant at the 1% level.

B.8 DV-Scaling and Non-Ceiling Robustness

The paper's baseline dependent variable is the judge-minus-panel deviation divided by the era-specific item-level standard deviation (Section 4.2). Table B7 reports the saturated triple-difference under five DV variants. Columns (1)–(4) re-estimate the saturated specification under four scalings: the era-specific item-level normalization (Col 1, baseline), the raw deviation in original score units (Col 2), the inverse-hyperbolic-sine transform (Col 3), and within-(item $\times$ season) standardization (Col 4). Column (5) replicates the baseline normalization on the non-ceiling PCS subsample (leave-one-out panel mean at most 9.0), stripping the ceiling-affected upper region. The architectural triple difference $\Delta^A = -.023$ is preserved on Col (1) and Col (5), confirming that the result is not a ceiling artifact. Cols (2)–(4) yield $\Delta^A > 0$ because raw PCS deviations span a wider score range than raw GOE deviations and the post-reform decline in raw PCS bias registers numerically larger than the corresponding GOE decline; the constituent post-2018 attenuation coefficients β_3 and β_4 are negative across all five scalings.

TABLE B7
DV-SCALING AND NON-CEILING ROBUSTNESS

	(1)	(2)	(3)	(4)	(5)
	Normalized	Raw level	asinh(dev)	Within-cell SD	Non-ceiling PCS
<i>Dep. var.: score deviation (alternative scalings)</i>					
SN × GOE	.093*** (.005)	.281*** (.015)	.249*** (.013)	.452*** (.023)	.093*** (.005)
SN × PCS	.035*** (.002)	.326*** (.014)	.305*** (.013)	.739*** (.031)	.035*** (.002)
SN × GOE × Post-2018	-.039*** (.005)	-.010 (.020)	-.019 (.017)	-.118*** (.027)	-.039*** (.005)
SN × PCS × Post-2018	-.016*** (.002)	-.135*** (.017)	-.124*** (.016)	-.255*** (.040)	-.016*** (.002)
SN × GOE × Post-2022	-.010*** (.004)	-.049** (.019)	-.042*** (.016)	-.050** (.024)	-.010*** (.004)
SN × PCS × Post-2022	-.002 (.002)	-.023 (.015)	-.023* (.014)	-.083** (.036)	-.002 (.002)
Item FE	Yes	Yes	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes	Yes	Yes
Δ^A (architectural)	-.023	.125	.106	.137	-.023
Δ^B (non-architectural)	.008	.026	.019	-.033	.008
Mean abs(DV)	.097	.483	.433	.769	.099
Observations	1,012,339	1,012,339	1,012,339	976,735	986,627
Judge clusters	543	543	543	543	543
Within R-squared	.010	.010	.011	.014	.010

Note.—All columns estimate the saturated triple-difference specification matching Table 3. Col (1) uses the paper’s baseline DV: the raw judge-minus-panel deviation divided by the era-specific item-level SD. Col (2) uses raw deviations in original score units; Col (3) applies the inverse-hyperbolic-sine transform; Col (4) standardizes by the within-(item × season) SD. Col (5) restricts PCS observations to leave-one-out panel mean ≤ 9.0 (GOE unchanged). Δ^A and Δ^B are the architectural and non-architectural triple differences (defined in Table 3). All models include item and judge fixed effects. Standard errors in parentheses, clustered by judge.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.